

Workshop - Week 7

1. For question (a), (c), (e), (g) : find the domain, intercepts, symmetry, slant asymptote, intervals of increase and decrease, local maximum and minimum values, concavity and points of inflection, and sketch the curve.

(a) $y = \frac{x^3+4}{x^2}$

- (b) Find the point on the curve $y = \sqrt{x}$ that is closest to the point (3, 0)

(c) $y = \frac{x^3}{(x+1)^2}$

- (d) Find the point on the ellipse $4x^2 + y^2 = 4$ that are the farthest away from the point (1, 0)

(e) $y = 1 + \frac{1}{2}x + e^{-x}$

- (f) Find the dimensions of the rectangle of largest area that can be inscribed in a circle of radius r

(g) $y = 1 - x + e^{1+x/3}$

- (h) Find an equation of the line through the point (3, 5) that cuts off the least area from the first quadrant.